Java Data Types

Java is statically typed and also a strongly typed language because, in Java, each type of data is predefined as part of the programming language and all constants or variables defined for a given program must be described with one of the Java data types.

Data Types in Java
Data types in Java are of different sizes and values that can be stored in the variable that is made as per convenience and circumstances to cover up all test cases.

Data types are mainly of two categories:

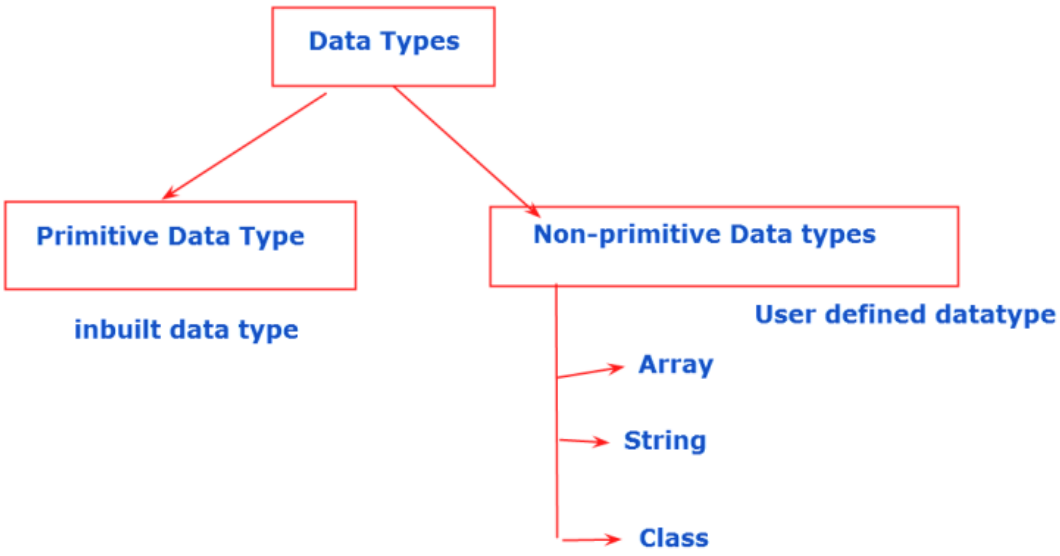
a. **Primitive Data Types**- These data types are already hard coded into the compiler to be recognized when the program is executed. Examples are- int,float etc.

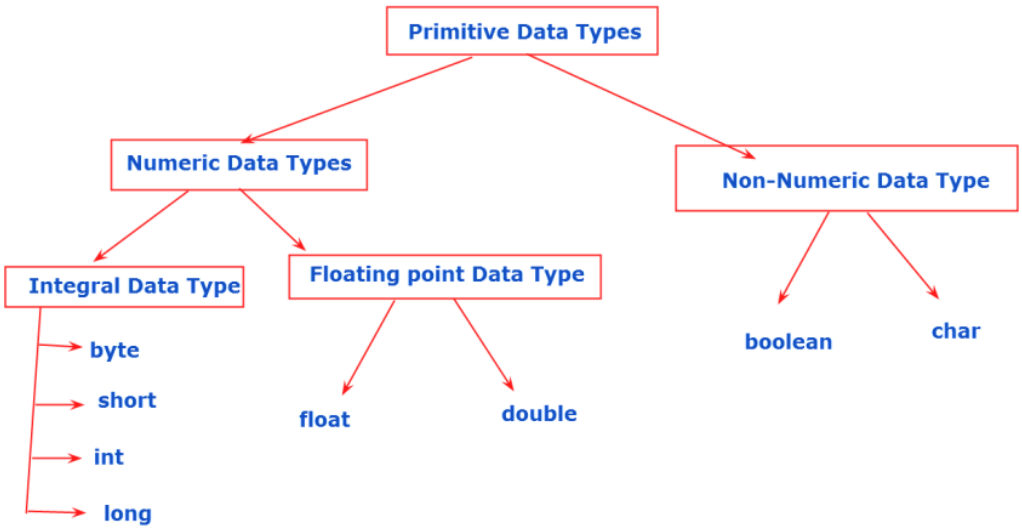
b. Non-Primitive Data Types- These data types are special types of data which are user defined, i,e, the program contains their definition. Some examples are- classes, interfaces etc.

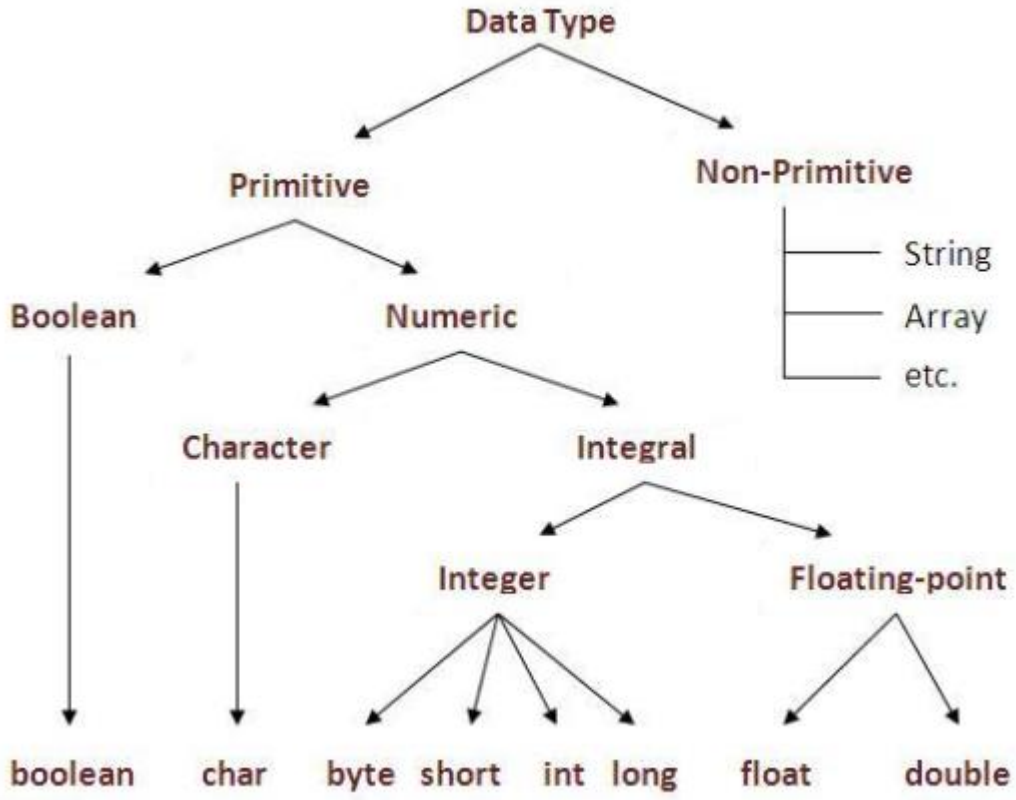
1. Primitive Data Types in Java

Java primitive data types are the ones which are predefined by the programming language which in this case is Java. Without primitive data types it would be impossible to frame programs. Primitive data types are also the building blocks of Non-primitive data types.

There are 8 types of Java primitive data types namely:
a. int
b. float
c. char
d. boolean
e. byte
f. short
g. long
h. double.


Data Types
Primitive Data Type
inbuilt data type
Non-primitive Data types
User defined datatype
Array
String
Class


Primitive Data Types
Numeric Data Types
Non-Numeric Data Type
Integral Data Type
Floating point Data Type
byte
short
int
long
float
double
boolean
char



Data Type	Size	Limit
byte	1 byte	-128 to 127
short	2 byte	-32,768 to 32,767
int	4 byte	-2,147,483,648 to 2,147,483,647
long	8 byte	-9,223,372,036,854,775,808 to 9,223,372,036,854,775,807
float	4 byte	6-7 significant decimal digits
double	8 byte	15 significant decimal digits
boolean	1 byte	true or false
char	2 byte	0 to 65,536 (unsigned)

Default value:

Below are the default values for all the data types:

Data Type	Default Value
byte	0
short	0
int	0
long	0
float	0.0f
double	0.0d
boolean	FALSE
char	\u0000'
String or Object	null

Below is the example to print default values of some of the variables:

```java
public class Byte {
    static byte byt;
    static long l;
    static float f ;
    static int i;
    static boolean b;
    static String s;

    public static void main(String[] args) {
        System.out.println("Byte = "+byt);
        System.out.println("long = "+i);
        System.out.println("float = "+i);
        System.out.println("int = "+i);
        System.out.println("boolean = "+b);
        System.out.println("String = "+s);
    }
}
```

```
Byte = 0
long = 0
float = 0
int = 0
boolean = false
String = null
```